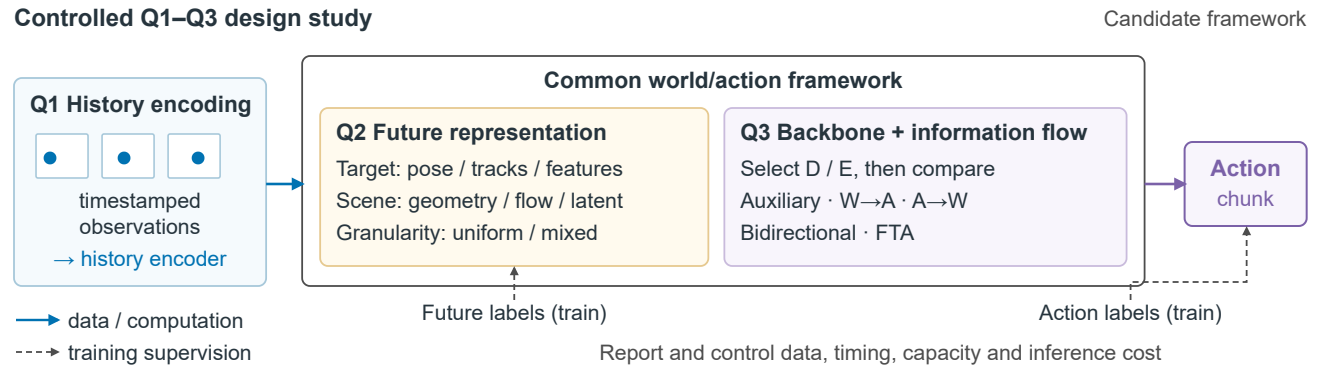
Controlled Q1–Q3 design study
Candidate framework
Q1 History encoding
timestamped observations
→ history encoder
Common world/action framework
Q2 Future representation
Target: pose / tracks / features
Scene: geometry / flow / latent
Granularity: uniform / mixed
Q3 Backbone + information flow
Select D / E, then compare
Auxiliary · W→A · A→W
Bidirectional · FTA
Action chunk
Future labels (train)
Action labels (train)
data / computation
training supervision
Report and control data, timing, capacity and inference cost